

# Technical Report

## Customer Churn Analysis & Predictive Retention Strategy

### 1. Introduction

#### 1.1 Background

Customer churn is one of the most important business challenges in subscription-based industries. High churn rates negatively impact profitability, customer lifetime value, and overall business growth.

This project aims to analyze customer churn behavior using data analytics and machine learning techniques to identify major churn drivers and recommend actionable retention strategies.

### 2. Project Objectives

The main objectives of this project were:

- Analyze customer churn behavior
- Identify important churn influencing factors
- Perform exploratory data analysis (EDA)
- Apply statistical and predictive analysis techniques
- Generate business recommendations for retention improvement

### 3. Dataset Description

The project used a customer churn dataset containing customer-level information such as:

- Customer tenure
- Monthly charges
- Total charges
- Contract type
- Payment methods
- Paperless billing status
- Senior citizen status
- Churn status

The dataset contained more than 500 rows and satisfied the project requirements.

### 4. Tools & Technologies Used

| Library      | Use                       |
|--------------|---------------------------|
| Numpy        | Analysis                  |
| Pandas       | Manipulation              |
| Matplotlib   | Arithmetic operations     |
| Seaborn      | Visualization             |
| Plotly       | Interactive visualization |
| Scikit-learn | Machine learning          |

### 5. Data Cleaning & Preparation

The following preprocessing steps were performed:

- Loaded raw dataset
- Checked dataset structure and datatypes
- Verified missing values
- Removed duplicate rows
- Verified target variable distribution
- Saved cleaned dataset for further analysis

No major missing value issues were identified in the dataset.

## 6. Exploratory Data Analysis (EDA)

EDA was performed to identify patterns, trends, and relationships within the data.

### 6.1 Churn Distribution

The churn analysis revealed:

- 89.4% retained customers
- 10.6% churned customers

This indicates relatively stable retention but still highlights business risk from customer loss.

### 6.2 Numerical Feature Analysis

Key numerical variables analyzed:

- Tenure
- Monthly Charges
- Total Charges

Visualizations used:

- Histograms
- Boxplots
- Correlation Heatmap

## 6.3 Categorical Feature Analysis

Categorical variables analyzed:

- Contract type
- Payment method
- Paperless billing

Visualizations:

- Count plots
- Churn comparison charts

# 7. Correlation Analysis

A correlation heatmap was generated to analyze relationships between variables.

## Major Findings

| Variable       | Correlation with Churn | Interpretation                         |
|----------------|------------------------|----------------------------------------|
| tenure         | -0.62                  | Strong negative relationship           |
| contract       | -0.18                  | Longer contracts reduce churn          |
| monthlyCharges | 0.05                   | Higher charges slightly increase churn |

The analysis identified customer tenure as the most significant churn predictor.

## 8. Predictive Modeling

### 8.1 Model Used

A Logistic Regression model was developed to predict customer churn probability.

### 8.2 Model Workflow

1. Feature selection
2. Train-test split
3. Model training
4. Prediction generation
5. Performance evaluation

## 9. Model Evaluation

The following evaluation techniques were used:

- Accuracy score
- Classification report
- Confusion matrix

The model successfully identified major churn patterns and customer risk factors.

## 10. Key Insights

1. Short-tenure customers are highly likely to churn.
2. Long-term contracts improve customer retention.
3. Customers with higher monthly charges show slightly higher churn risk.
4. Contract structure plays a major role in customer retention behavior.

# **11. Business Recommendations**

## **Recommendation 1 — Loyalty Programs**

Introduce onboarding and loyalty programs for new customers.

## **Recommendation 2 — Long-Term Contract Incentives**

Provide discounts and benefits for yearly subscriptions.

## **Recommendation 3 — Customer Retention Campaigns**

Target high-risk customers using predictive analytics.

## **Recommendation 4 — Pricing Optimization**

Evaluate pricing strategies for high monthly charge customers.

# **12. Conclusion**

This project successfully analyzed customer churn behavior using business analytics and machine learning techniques.

The findings demonstrated that customer tenure is the strongest indicator of churn, while long-term contracts improve retention rates. Predictive modeling further enabled the identification of high-risk customer groups.

The proposed recommendations can help businesses reduce churn, improve customer satisfaction, and strengthen long-term revenue stability.